WEEK 3 MINI PROJECT

Serverless Contact Form

Using AWS Lambda & API Gateway

Submitted by	Aditya Shukla
Cloud Platform	Amazon Web Services (AWS)
Services Used	AWS Lambda, Amazon API Gateway, Postman
Region	Europe (Stockholm) — eu-north-1

Project objective: Build and test a simple serverless API that accepts contact-form data and returns a JSON response.

1. Project Overview

This mini project demonstrates a serverless contact-form backend using AWS Lambda and Amazon API Gateway. The Lambda function receives JSON input, validates the required fields, logs the submitted payload for demonstration, and returns a JSON success or error response. API Gateway exposes the function through an HTTP endpoint.

2. Objectives

• Create an AWS Lambda function named **contact-form-handler**.

• Integrate the function with an HTTP API using Amazon API Gateway.

• Accept contact-form fields: name, email, and message.

• Test the endpoint with a POST request in Postman and verify the response.

3. AWS Services and Tools

Service / Tool	Purpose
AWS Lambda	Runs the Python handler without managing servers.
Amazon API Gateway (HTTP API)	Provides an HTTP endpoint and forwards requests to Lambda.
Postman	Sends a POST request and displays the HTTP response.

4. Implementation Procedure

Step 1 — Create Lambda function

Created a Python-based Lambda function named **contact-form-handler** using the AWS Lambda console.

Step 2 — Add handler code

Added Python code to read the incoming JSON body, validate name/email/message, log the submitted data, and return a JSON response.

Step 3 — Configure API Gateway

Added an API Gateway HTTP API trigger to expose the Lambda function through an endpoint. The API trigger shows authorization as NONE, so the demo endpoint is publicly callable.

Step 4 — Test with Postman

Sent a POST request with a JSON payload and confirmed that the endpoint returned **200 OK** and **success: true**.

5. Test Request

HTTP method: **POST**

Request body (JSON):

```
{
  "name": "Aditya",
  "email": "aditya@example.com",
  "message": "Testing my serverless contact form"
}
```

6. Test Result

The Postman test returned **200 OK**. The JSON response contained **success: true**, the message “Contact form submitted successfully”, and the same name, email, and message values sent in the request.

7. Screenshots / Evidence

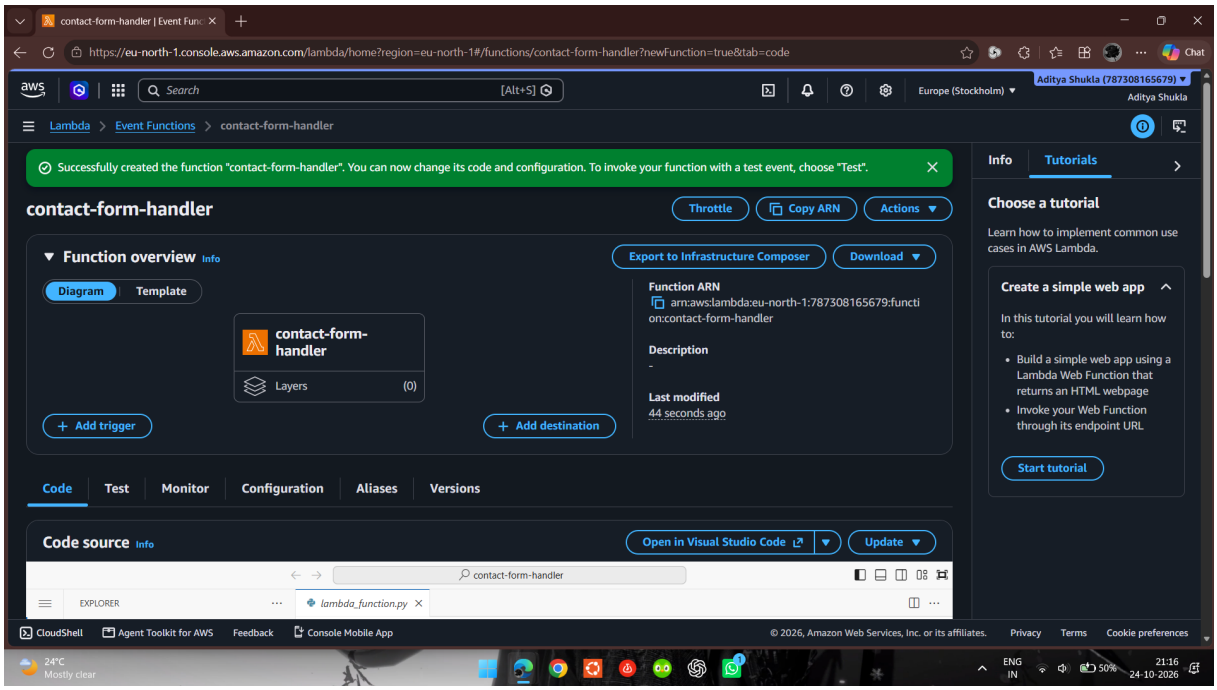


Figure 1. AWS Lambda function created: contact-form-handler.

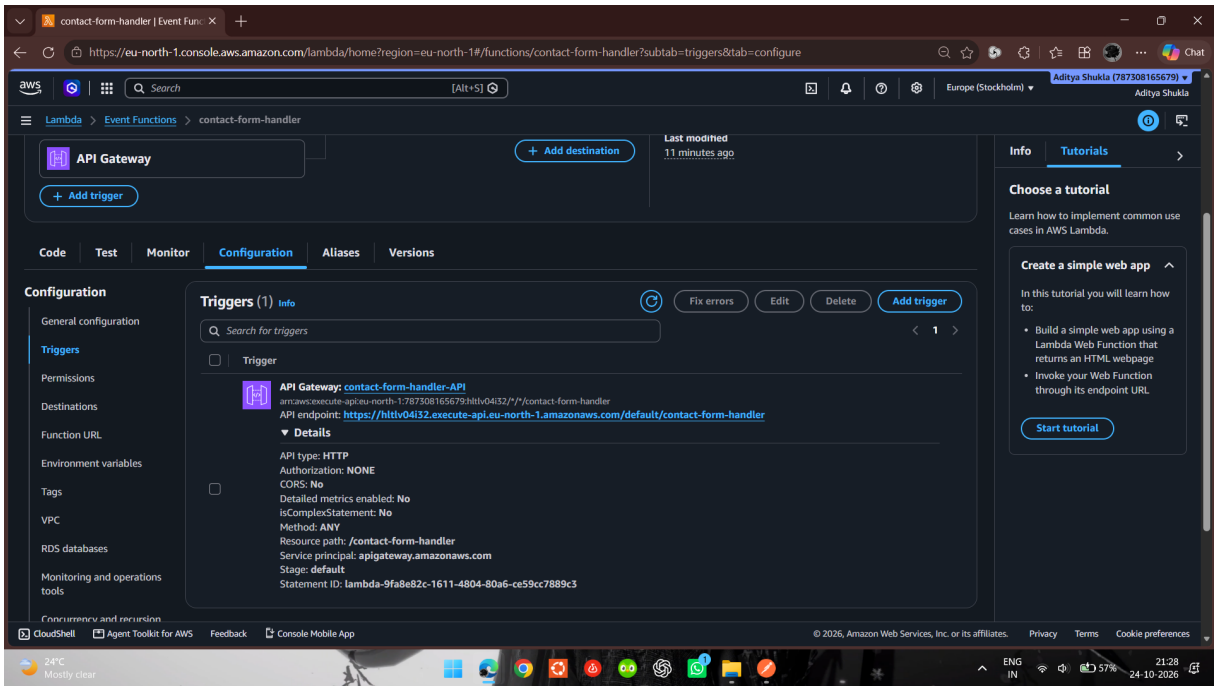


Figure 2. API Gateway HTTP API trigger integrated with the Lambda function.

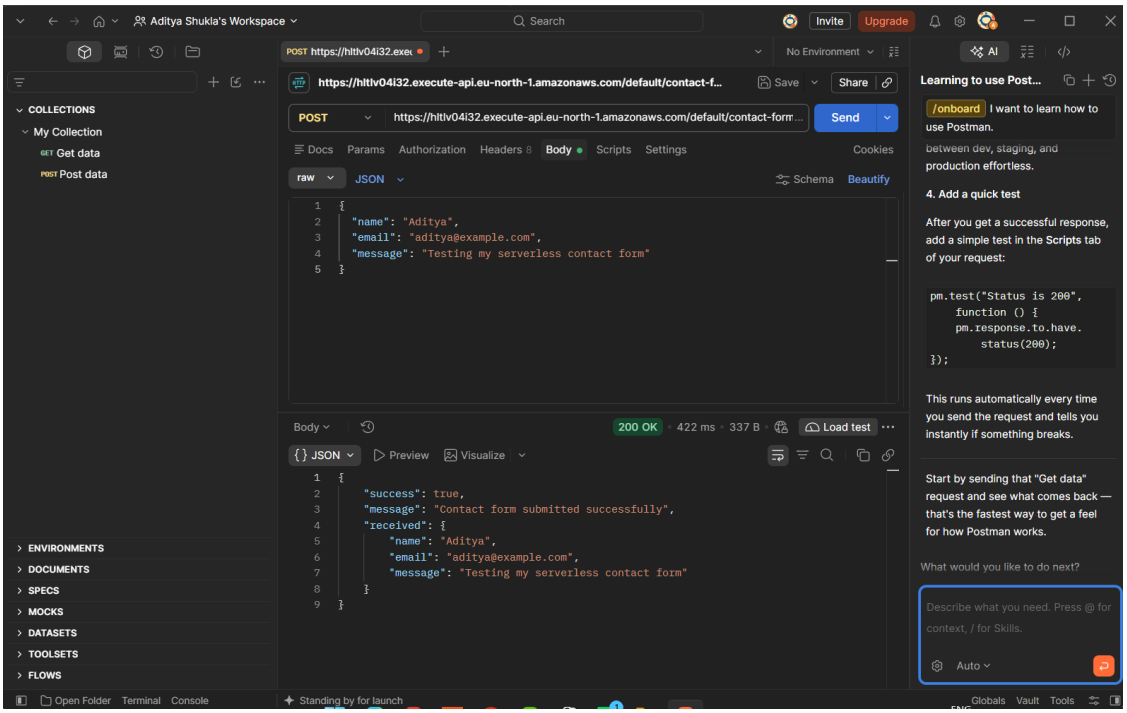


Figure 3. Postman POST request returns 200 OK and a successful JSON response.

8. Conclusion

The serverless contact-form API was created and tested successfully. AWS Lambda handled the request processing, API Gateway exposed the function through an HTTP endpoint, and Postman confirmed the successful response. This implementation demonstrates request validation and response handling; it does not persist submissions to a database or send email notifications.

Security note

The API trigger is configured with Authorization: NONE for this learning demo. Do not use this open configuration for sensitive or production contact-form data. Add suitable authorization, throttling, validation, and monitoring before production use.